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$$\frac{\overline{P}}{\overline{E}}_{Gi} = e^{\frac{\ln(22.29\times15.54\ldots\times10.34\times8.35)}{10}} = e^{\frac{ln(38,016,128,040)}{10}} = e^{24.3613/10} = 11.4287.$$

3. Calculate the harmonic mean P/E for the 10 stocks.

Solution:

The harmonic mean is calculated as:

$$\overline{X}_H = \frac{n}{\sum_{i=1}^{n}(1/X_i)},$$

$$\overline{X}_H = \frac{10}{\left(\frac{1}{22.29}\right)+\left(\frac{1}{15.54}\right)+\ldots+\left(\frac{1}{10.34}\right)+\left(\frac{1}{8.35}\right)},$$

$$\overline{X}_H = 10/0.9247 = 10.8142.$$

In finance, the weighted harmonic mean is used when averaging rates and other multiples, such as the P/E ratio, because the harmonic mean gives equal weight to each data point, and reduces the influence of outliers. These calculations can be performed using Excel:

- To calculate the arithmetic mean or average return, the =AVERAGE(return1, return2, …) function can be used.
- To calculate the geometric mean return, the =GEOMEAN(return1, return2, …) function can be used.
- To calculate the harmonic mean return, the =HARMEAN(return1, return2, …) function can be used.

In addition to arithmetic, geometric, and harmonic means, two other types of means can be used. Both the trimmed and the winsorized means seek to minimize the impact of outliers in a dataset. Specifically, the **trimmed mean** removes a small defined percentage of the largest and smallest values from a dataset containing our observation before calculating the mean by averaging the remaining observations.

A winsorized mean replaces the extreme observations in a dataset to limit the effect of the outliers on the calculations. The **winsorized mean** is calculated after replacing extreme values at both ends with the values of their nearest observations, and then calculating the mean by averaging the remaining observations.

However, the key question is: Which mean to use in what circumstances? The choice of which mean to use depends on many factors, as we describe in Exhibit 8:

- Are there outliers that we want to include?
- Is the distribution symmetric?
- Is there compounding?
- Are there extreme outliers?